

Angular Analysis of

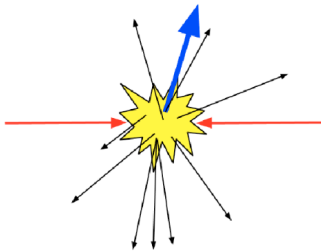
$B^0 \rightarrow K^{*0}(\rightarrow K^+\pi^-)\mu^+\mu^-$ Decays

Giuseppe Luciano — 28 May 2026

FCNC Rare Decays · LHCb · Angular Observables · New Physics

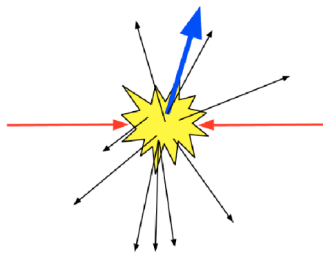
Where do we look for New Physics?

Direct search.

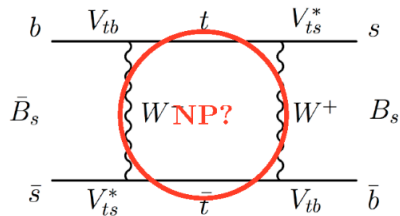


Where do we look for New Physics?

Direct search.

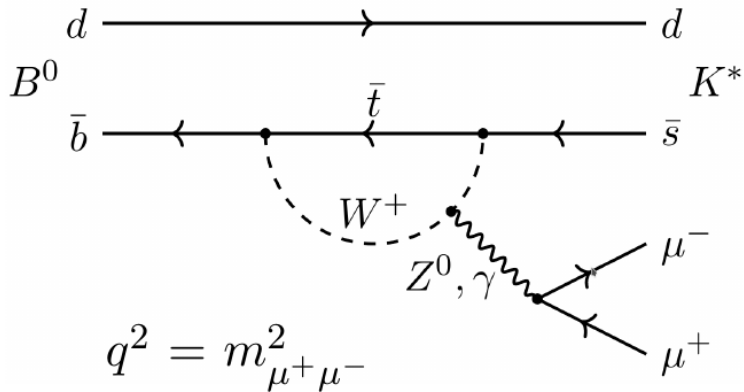


Indirect search.



History: top quark mass predicted by quark mixing

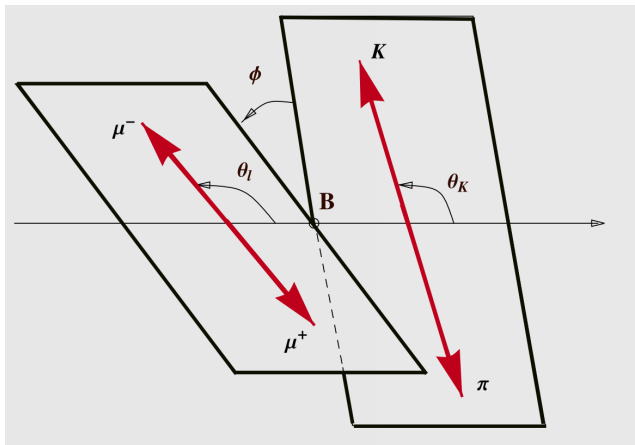
The Decay Channel



The Decay $B^0 \rightarrow K^{*0}(\rightarrow K\pi)\mu^+\mu^-$

Kinematic variables

- q^2 — dimuon invariant mass squared
 - θ_ℓ — μ^+ angle in dimuon rest frame
 - θ_K — K^+ angle in K^{*0} rest frame
 - ϕ — angle between decay planes
- $$\Omega = (\cos \theta_\ell, \cos \theta_K, \phi)$$



Differential decay width

The differential decay width is given by:

$$\frac{d^4\Gamma(\bar{B}^0 \rightarrow \bar{K}^{*0} \mu^+ \mu^-)}{dq^2 d\Omega} \propto \sum_i l_i(q^2) f_i(\Omega) \quad (1)$$

$$\frac{d^4\bar{\Gamma}(B^0 \rightarrow K^{*0} \mu^+ \mu^-)}{dq^2 d\Omega} \propto \sum_i \bar{l}_i(q^2) f_i(\Omega) \quad (2)$$

- ▶ l_i and $\bar{l}_i$ are functions of the form factor and coupling.
- ▶ $f_{ij}(\vec{\Omega})$ are known functions of the angular variables.

$$S_i(q^2) = \frac{l_i + \bar{l}_i}{\frac{d\Gamma}{dq^2} + \frac{d\bar{\Gamma}}{dq^2}}$$

- ▶ $F_L = \frac{1}{2}(1 - S_2)$ is the fraction of longitudinally polarized K^* .
- ▶ $A_{FB} = \frac{3}{4}S_6$ is the forward-backward asymmetry of the leptons.

Differential decay width

$$\frac{1}{d(\Gamma + \bar{\Gamma})/dq^2} \frac{d^4(\Gamma + \bar{\Gamma})}{dq^2 d\Omega} = \frac{9}{32\pi} \left[\frac{3}{4}(1 - F_L(q^2))\sin^2 \theta_K + F_L(q^2)\cos^2 \theta_K \right. \\ \left. + \frac{1}{4}(1 - F_L(q^2))\sin^2 \theta_K \cos 2\theta_\ell \right. \quad (3)$$

$$+ S_3(q^2)\sin^2 \theta_K \sin^2 \theta_\ell \cos 2\phi$$

$$+ S_4(q^2)\sin 2\theta_K \sin 2\theta_\ell \cos \phi$$

$$+ S_5(q^2)\sin 2\theta_K \sin \theta_\ell \cos \phi$$

$$+ \frac{4}{3}A_{FB}(q^2)\sin^2 \theta_K \cos \theta_\ell$$

$$+ S_7(q^2)\sin 2\theta_K \sin \theta_\ell \sin \phi$$

$$+ S_8(q^2)\sin 2\theta_K \sin 2\theta_\ell \sin \phi$$

$$+ \underbrace{S_9(q^2)}_{\text{Fit target}} \underbrace{\sin^2 \theta_K \sin^2 \theta_\ell \sin 2\phi}_{\text{Angular dependence}} \Big]. \quad (4)$$

Optimized Observables

$$P'_1 = \frac{2S_3}{1 - F_L},$$

$$P'_2 = \frac{2}{3} \frac{A_{FB}}{1 - F_L},$$

$$P'_3 = -\frac{S_9}{1 - F_L}$$

$$P'_4 = \frac{S_4}{\sqrt{F_L(1 - F_L)}},$$

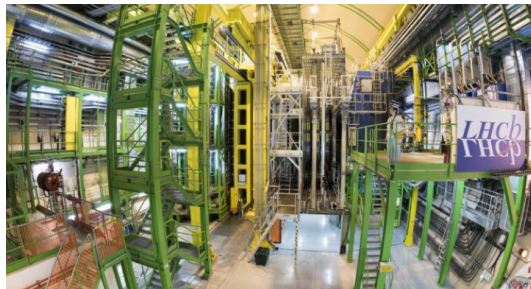
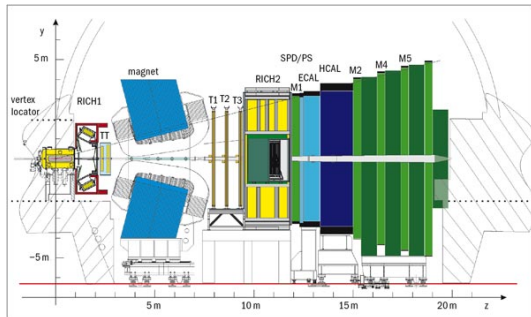
$$P'_5 = \frac{S_5}{\sqrt{F_L(1 - F_L)}},$$

$$P'_6 = \frac{S_7}{\sqrt{F_L(1 - F_L)}},$$

$$P'_8 = \frac{S_8}{\sqrt{F_L(1 - F_L)}}.$$

P_i does not depend on the form factors at leading order in α_s and $1/m_b$.

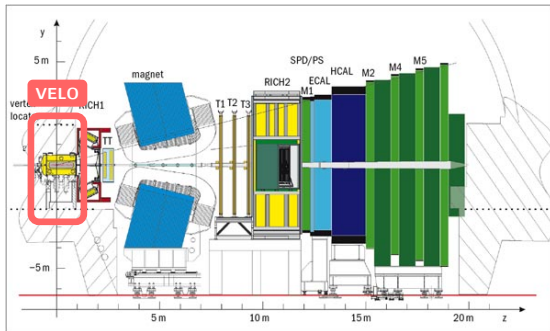
The LHCb Detector



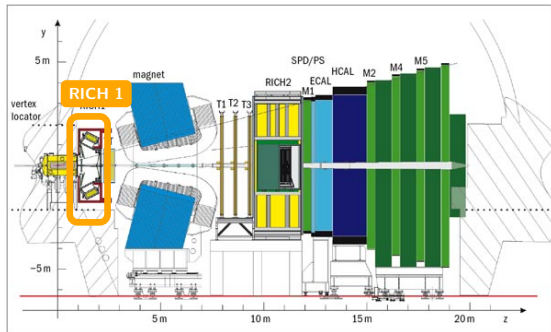
LHCb Detector — Subsystems

VELO — Vertex Locator

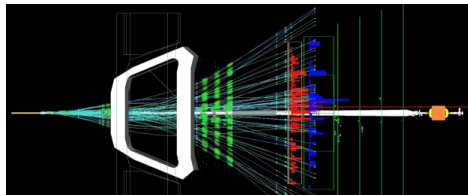
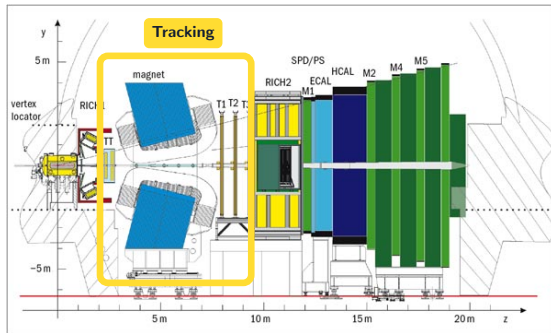
- ▶ Silicon strip detector.
- ▶ Impact parameter resolution $\sim 20 \mu\text{m}$



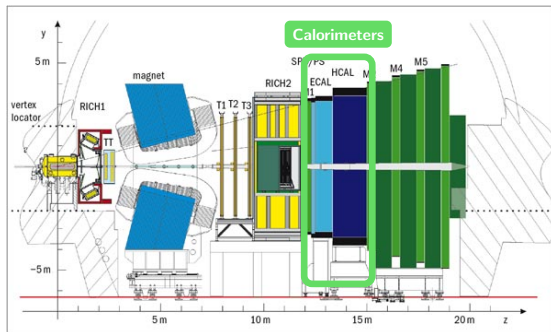
LHCb Detector — Subsystems



LHCb Detector — Subsystems



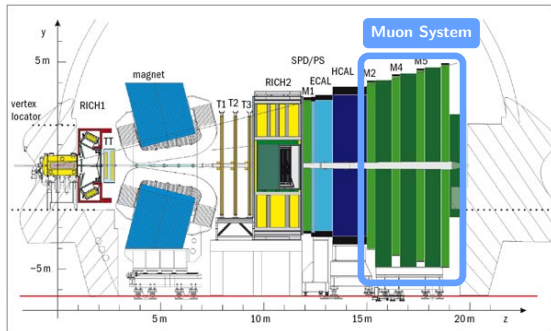
LHCb Detector — Subsystems



Calorimeters

- ▶ ECAL measures electromagnetic energy
- ▶ HCAL measures hadronic energy
- ▶ Used in the hardware trigger

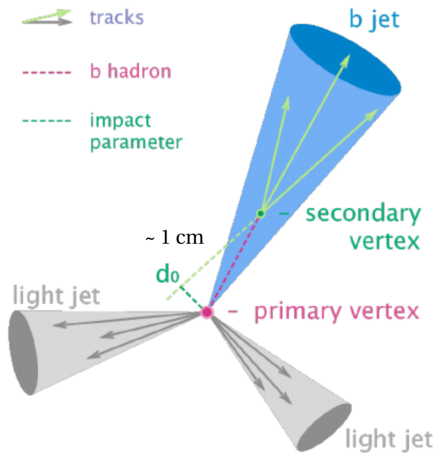
LHCb Detector — Subsystems



Muon System

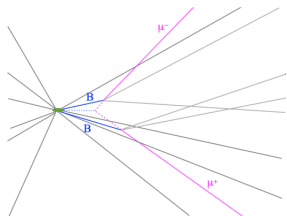
- Five muon stations, M1–M5
- Multi-wire proportional chambers
- Key subsystem for the $\mu^+\mu^-$ final state
- Triggers on high- p_T muons

Data Sample



Background Suppression

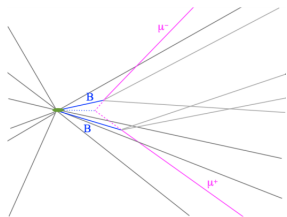
Combinatorial background,



BDT: retains $> 85\%$ signal
rejects $> 97\%$ background

Background Suppression

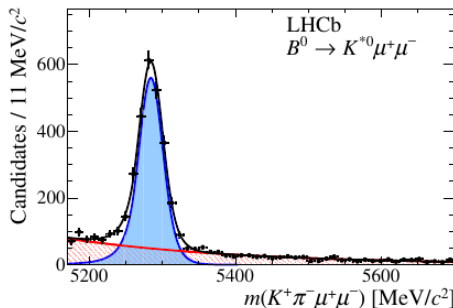
Combinatorial background,



BDT: retains $> 85\%$ signal
rejects $> 97\%$ background

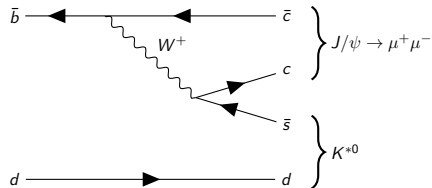
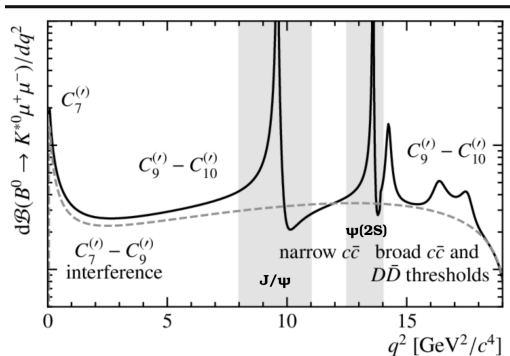
Peaking backgrounds,

$$\begin{aligned}\Lambda_b^0 &\rightarrow p K^- \mu^+ \mu^- \\ B_s^0 &\rightarrow \phi(\rightarrow K^+ K^-) \mu^+ \mu^- \end{aligned} \quad (5)$$



Residual $< 1\%$.

Resonant Contributions



Removed regions where $8.0 < q^2 < 11.0 \text{ GeV}^2/c^4$ for the J/ψ ,
 $12.5 < q^2 < 15.0 \text{ GeV}^2/c^4$ for the $\psi(2S)$.

S-wave Contributions

The $K^{*0}(892)$ resonance is a spin-1 state $\rightarrow$ **P-wave contribution**.

$K_0^*(800)_{m=845\text{MeV}, \Gamma=468\text{MeV}}$ **and** $K_0^*(1410)_{m=1425\text{MeV}, \Gamma=270\text{MeV}}$ system in spin-0 state $\rightarrow$ **S-wave**.

S-wave Contributions

The $K^{*0}(892)$ resonance is a spin-1 state $\rightarrow$ **P-wave contribution**.

$K_0^*(800)_{m=845\text{MeV}, \Gamma=468\text{MeV}}$ **and** $K_0^*(1410)_{m=1425\text{MeV}, \Gamma=270\text{MeV}}$ system in spin-0 state $\rightarrow$ **S-wave**.

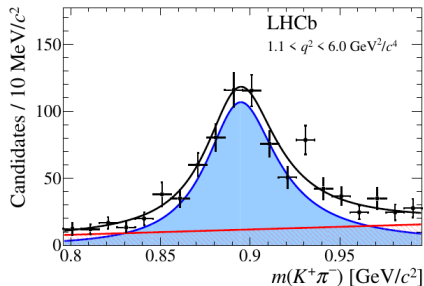
$$\begin{aligned} \frac{1}{d(\Gamma + \bar{\Gamma})/dq^2} \frac{d^4(\Gamma + \bar{\Gamma})}{dq^2 d\vec{\Omega}} \Big|_{S+P} &= (1 - F_S) \frac{1}{d(\Gamma + \bar{\Gamma})/dq^2} \frac{d^4(\Gamma + \bar{\Gamma})}{dq^2 d\vec{\Omega}} \Big|_P \\ &+ \frac{3}{16\pi} F_S \sin^2 \theta_\ell \\ &+ \frac{9}{32\pi} (S_{11} + S_{13} \cos 2\theta_\ell) \cos \theta_K \\ &+ \frac{9}{32\pi} (S_{14} \sin 2\theta_\ell + S_{15} \sin \theta_\ell) \sin \theta_K \cos \phi \\ &+ \frac{9}{32\pi} (S_{16} \sin \theta_\ell + S_{17} \sin 2\theta_\ell) \sin \theta_K \sin \phi \end{aligned}$$

S-wave Contributions

The $K^{*0}(892)$ resonance is a spin-1 state $\rightarrow$ **P-wave contribution**.

$K_0^*(800)_{m=845\text{MeV}, \Gamma=468\text{MeV}}$ and $K_0^*(1410)_{m=1425\text{MeV}, \Gamma=270\text{MeV}}$ system in spin-0 state $\rightarrow$ **S-wave**.

$$\frac{1}{d(\Gamma + \bar{\Gamma})/dq^2} \frac{d^4(\Gamma + \bar{\Gamma})}{dq^2 d\vec{\Omega}} \Big|_{S+P} = (1 - F_S) \frac{1}{d(\Gamma + \bar{\Gamma})/dq^2} \frac{d^4(\Gamma + \bar{\Gamma})}{dq^2 d\vec{\Omega}} \Big|_P$$

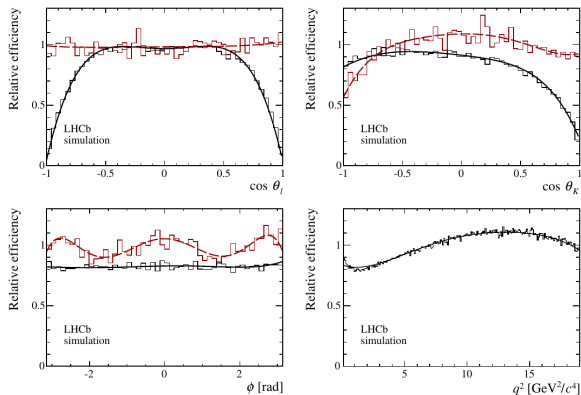


$$\begin{aligned} & + \frac{3}{16\pi} F_S \sin^2 \theta_\ell \\ & + \frac{9}{32\pi} (S_{11} + S_{13} \cos 2\theta_\ell) \cos \theta_K \\ & + \frac{9}{32\pi} (S_{14} \sin 2\theta_\ell + S_{15} \sin \theta_\ell) \sin \theta_K \cos \phi \\ & + \frac{9}{32\pi} (S_{16} \sin \theta_\ell + S_{17} \sin 2\theta_\ell) \sin \theta_K \sin \phi \end{aligned}$$

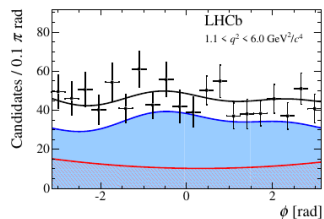
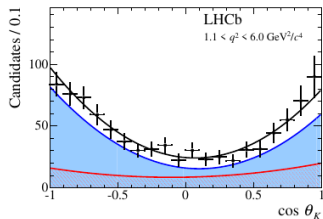
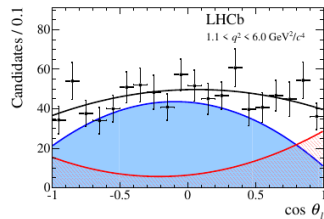
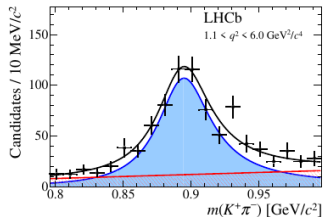
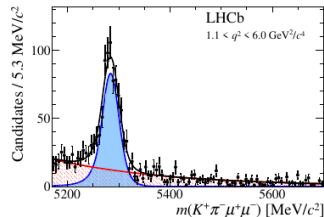
Acceptance Correction

$$\epsilon(\cos \theta_\ell, \cos \theta_K, \phi, q^2) = \sum_{ijmn} C_{ijmn} L_i(\cos \theta_\ell) L_j(\cos \theta_K) L_m(\phi) L_n(q^2) \quad (6)$$

- Black solid line: region $0.1 < q^2 < 0.98 \text{ GeV}^2/c^4$.
- Red solid line: region $18 < q^2 < 19 \text{ GeV}^2/c^4$.

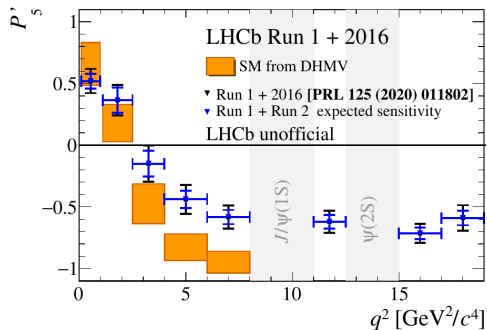


Unbinned Maximum Likelihood Fit



Results

When we look at $P'_5 = S_5/\sqrt{F_L(1-F_L)}$ in the bins $4.0 < q^2 < 6.0$ and $6.0 < q^2 < 8.0 \text{ GeV}^2/c^4$, the local discrepancies are 2.8 and 3.0σ respectively.



Systematic Uncertainties

Source	F_L	S_3-S_9	A_3-A_9	$P_1-P'_8$	q_0^2 GeV ² /c ⁴
Acceptance stat. uncertainty	< 0.01	< 0.01	< 0.01	< 0.01	0.01
Acceptance polynomial order	< 0.01	< 0.02	< 0.02	< 0.04	0.01–0.03
Data-simulation differences	0.01–0.02	< 0.01	< 0.01	< 0.01	< 0.02
Acceptance variation with q^2	< 0.01	< 0.01	< 0.01	< 0.01	–
$m(K^+\pi^-)$ model	< 0.01	< 0.01	< 0.01	< 0.03	< 0.01
Background model	< 0.01	< 0.01	< 0.01	< 0.02	0.01–0.05
Peaking backgrounds	< 0.01	< 0.01	< 0.01	< 0.01	0.01–0.04
$m(K^+\pi^-\mu^+\mu^-)$ model	< 0.01	< 0.01	< 0.01	< 0.02	< 0.01
Det. and prod. asymmetries	–	–	< 0.01	< 0.02	–

For the P'_i observables, statistical uncertainties are typically at the 10-30% level of the measured value, while systematic uncertainties are usually only 10-30% of the statistical uncertainty.

Theoretical Framework

The effective Hamiltonian is often used to describe the decay:

$$\mathcal{H}_{\text{eff}} \propto \sum_i C_i(q^2) \mathcal{O}_i$$

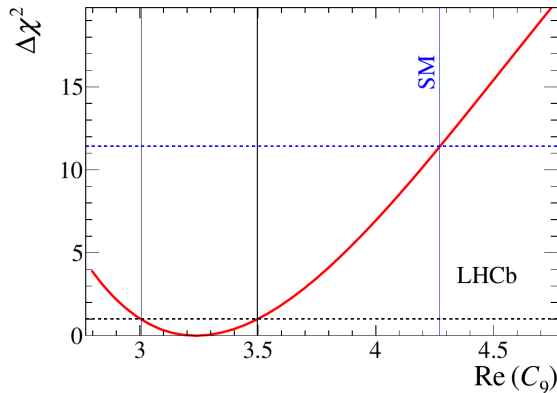
- ▶ $C_i \rightarrow$ Wilson coefficients
- ▶ $\mathcal{O}_i \rightarrow$ Operators (e.g., $\mathcal{O}_9$, vector current; $\mathcal{O}_{10}$, axial-vector current)

Global Fit to Wilson Coefficients

It is possible to explain the tensions in P'_5 by a shift in the Wilson coefficient C_9 :

$$\Delta\text{Re}(C_9) = -0.99^{+0.25}_{-0.21}$$

- ▶ Negative shift $\Rightarrow$ effective vector coupling suppressed relative to SM
- ▶ $\sim 3.4\sigma$ from the SM expectation $\Delta\text{Re}(C_9) = 0$



Possible Explanations

Charm loops can modify the Wilson coefficients, and possibly explain the observed tensions in P'_5 .

BUT...

Ciuchini et al.

Corrections not under sufficient theoretical control, too close to J/ψ resonance.

Could mimic NP.

Possible Explanations

Charm loops can modify the Wilson coefficients, and possibly explain the observed tensions in P'_5 .

BUT...

Ciuchini et al.

Corrections not under sufficient theoretical control, too close to J/ψ resonance.
Could mimic NP.

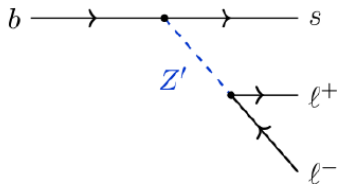
Descotes-Genon et al.

These corrections are well-constrained by other channels and **cannot plausibly explain the full shift.**

BSM Explanations

Introduction of a new massive neutral vector boson (Z')

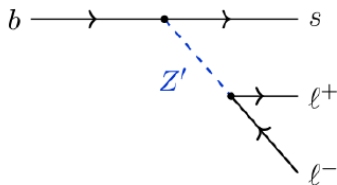
New heavy gauge bosons



BSM Explanations

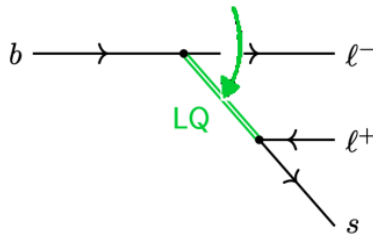
Introduction of a new massive neutral vector boson (Z')

New heavy gauge bosons



Leptoquarks are hypothetical particles that can directly couple quarks to leptons.

observables are altered by new (virtual) particles



Conclusions

Summary

- ✓ Full angular analysis of $B^0 \rightarrow K^{*0} \mu^+ \mu^-$
- ✓ Extraction of the optimized P'_i observables
- ✓ Local tensions in P'_5 at 2.8 and 3.0σ
- ✓ Best-fit Wilson-coefficient shift:
 $\Delta \text{Re}(C_9) = -0.99^{+0.25}_{-0.21}$

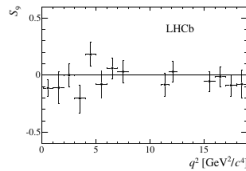
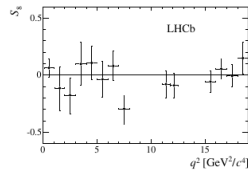
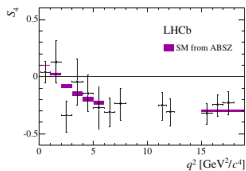
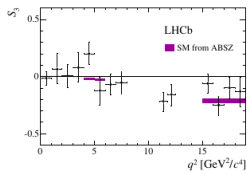
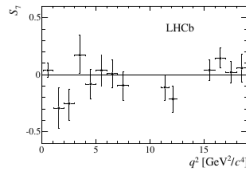
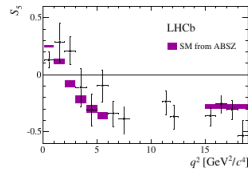
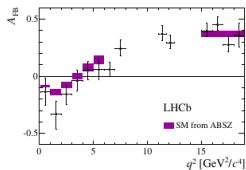
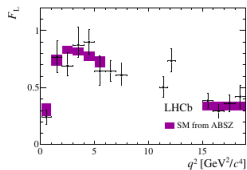
Open questions

Hadronic charm-loop effects?
 Z' boson or other physics?
Statistical fluctuation?

More data will be decisive.

Thanks for your attention!

Backup



Backup

